

Education

Brown University

Ph.D., Cognitive Science (GPA: 4.0/4.0)

Advisors: Dr. William C. Heindel, Dr. Elena K. Festa

Providence, RI

2021 – 2026

University of British Columbia (UBC)

B.A., Cognitive Systems (Class 1 Standing)

Advisors: Dr. Ronald A. Rensink, Dr. Karon E. MacLean

Vancouver, BC

2020

Professional Appointment

Medical College of Wisconsin

Postdoctoral Researcher

Advisor: Dr. Iftikhar J. Kullo

Milwaukee, WI

07/2026 – Present

Publications

1. **Ip, J.**, Heindel, W. C., Ott, B. R., Festa, E. K. (2026). Distinct theta and alpha electrophysiological dynamics during audiovisual semantic processing in healthy aging and mild cognitive impairment. *Aging Brain*, 10, 100163. <https://doi.org/10.1016/j.nbas.2026.100163>
2. Seifi, H., Fazlollahi, F., Oppermann, M., Sastrillo, J.A., **Ip, J.**, Agrawal, A., Park, G., Kuchenbecker, K.J., & MacLean, K.E. (2019). Haptipedia: Accelerating Haptic Device Discovery to Support Interaction and Engineering Design. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, CHI '19*, 1-12. <https://doi.org/10.1145/3290605.3300788>
Manuscripts In Preparation
3. **Ip, J.**, Heindel, W.C., & Festa, E.K. (*In Prep*). Cortical Spectral Dynamics of Cross- and Within-Stream Visual Search.
4. **Ip, J.**, Heindel, W.C., & Festa, E.K. (*In Prep*). Cortical Spectral Dynamics of Conjunctive Feature Binding in Memory.

Talks, Conference Presentations & Workshops

1. **Ip, J.**, Heindel, W.C., & Festa, E.K. (2026). Theta and Alpha EEG Dynamics Reveal Altered Semantic Processing in Aging and MCI [Poster Presentation]. International Neuropsychological Society. Philadelphia, PA.
2. **Ip, J.** (2025). Alpha/Theta Characterization of Semantic Processing Dynamics in Healthy Aging and Mild Cognitive Impairment [Talk]. Cognitive and Psychological Sciences Social/Cognitive Seminar, Brown University. Providence, Rhode Island.
3. **Ip, J.** (2025). Agentic AI Workflows in AWS HealthOmics: A Multiomics Example [Talk]. Healthcare and Life Sciences Technical Forum, Amazon Web Services. Seattle, Washington.
4. **Ip, J.**, Heindel, W.C., & Festa, E.K. (2024). Differential Attentional Costs of Encoding Single- vs. Multiple-Feature Objects in Visual Working Memory: An ERP Study [Poster Presentation]. Object Perception, Visual Attention, and Visual Memory Conference. New York City, New York.
5. **Ip, J.**, Heindel, W.C., & Festa, E.K. (2024). Differential EEG Markers of Selective Attention and Feature Binding in Visual Search. [Poster Presentation, VSS 2024]. *Journal of Vision*, 24(10), 1066. <https://doi.org/10.1167/jov.24.10.1066>
6. **Ip, J.**, Heindel, W.C., & Festa, E.K. (2023). Age-Related Effects of Audiovisual Semantic Congruency on Living and Nonliving Object Perception [Poster Presentation]. Cognitive Neuroscience Society (CNS) 30th Annual Meeting. San Francisco, California.
7. **Ip, J.**, Chin, N., & Rensink, R. (2021). Correlation Perception in Scatterplots is Invariant to Dot Size. [Poster Presentation, VSS 2021]. *Journal of Vision*, 21(9), 2171. <https://doi.org/10.1167/jov.21.9.2171>

8. Seifi, H., **Ip, J.**, Agrawal, A., Kuchenbecker, K. J., & MacLean, K. E. (2019). Toward Expert-Sourcing of a Haptic Device Repository. *CHI Workshop on Crowds and Creativity*, 1-4.
9. Seifi, H., Fazlollahi, F., Oppermann, M., Sastrillo, J. A., **Ip, J.**, Agrawal, A., Tembo, T., Park, G., Kuchenbecker, K. J., & MacLean, K. M. (2019). Haptipedia: A Haptic Device Library to Support Interaction and Engineering Design [Poster Presentation]. DFP Design Showcase. Vancouver, Canada.
10. **Ip, J.***, Pertels, Y.*, Chai, W.*, & Thongprasert, S*. (2017). Image Transitions: Visual Search in the Dynamic World [Poster Presentation]. UBC Multidisciplinary Undergraduate Research Conference. Vancouver, Canada.

*Denotes equal contribution.

Selected Awards & Honors

Sigma Xi , Scientific Research Honor Society	2024
Manning Graduate Fellowship IV , Brown University	2022–2023
Conference Travel Fund , Brown University	2023, 2024, 2026
Elsevier/Vision Research Virtual Travel Award , Vision Sciences Society	2021
Undergraduate Student Research Award in Computer Science , NSERC (\$4,500)	2019

Research Experience

Department of Cognitive and Psychological Sciences, Brown University Ph.D. Student/Candidate <i>Topics:</i> Cognitive Neuroscience, Sensory Perception, Memory, Aging, EEG	Providence, RI 08/2021 – 05/2026
Visual Cognition Lab, UBC Psychology Project Leader and Research Assistant with Dr. Ronald A. Rensink <i>Topics:</i> Information Visualization, Vision Science, Psychophysics	Vancouver, BC 09/2016 – 04/2021
UBC Computer Science & Max Planck Institute for Intelligent Systems Research Assistant with Dr. Karon E. MacLean and Dr. Hasti Seifi <i>Topics:</i> Human-Computer Interaction, Haptics, Crowdsourcing	Vancouver, BC 01/2018 – 08/2019

Teaching Experience

Research Mentor , Brown University Mentored and trained 4 undergraduate students on EEG and behavioral research methods.	Providence, RI 2024 – 2025
Graduate Teaching Assistant , Brown University CLPS 0702: Introduction to Clinical Psychology CLPS 1702: Psychological Assessment CLPS 1900: Research Methods and Design CLPS 0010: Mind, Brain and Behavior (Head TA) CLPS 1420: Cognitive Neuropsychology CLPS 0010: Mind, Brain and Behavior Led weekly sections, hosted weekly office hours, graded research papers and presentations, and managed teams of undergraduate and graduate TAs.	Providence, RI 01/2025 – 05/2025 09/2024 – 12/2024 01/2024 – 05/2024 09/2023 – 12/2023 01/2023 – 05/2023 09/2022 – 12/2022
Directed Studies Supervisor , UBC COGS 402: Research in Cognitive Systems Supervised a senior thesis project in vision science. Mentored and monitored student progress on experimental design, coding, data analysis, and paper writing.	Vancouver, BC 01/2020 – 04/2020
Workshop Instructor , UBC Visual Cognition Lab	Vancouver, BC

Workshop on Psychophysics and Data Analysis Methods 03/2019
Designed and taught four workshop sessions on psychophysics methods.

Undergraduate Teaching Assistant, UBC Vancouver, BC
COGS 303: Research Methods in Cognitive Systems 09/2018 – 12/2018
Graded weekly written assignments, quizzes, in-class activities, and final research reports while providing academic support for 39 students through email, weekly office hours, and exam review sessions.

Other Professional Experience

Amazon Web Services (AWS) Seattle, WA
Technical Writer Intern 06/2025 – 08/2025
Wrote and published API documentation and code samples for AWS HealthOmics, a cloud computing platform for large-scale bioinformatics analyses in precision medicine. Tested cloud-based agentic AI workflows for a neuroscientific use case. Presented a live demonstration and proof-of-concept findings at the healthcare and life sciences internal forum (50+ in attendance).
Technology: Amazon S3, EC2, IAM, CloudWatch, Docker, Git, VSCode, agentic AI and LLM prompting (Claude Sonnet 3.7)

Advesa Digital Solutions Inc. Burnaby, BC
Technical Writer 06/2020 – 06/2021
Wrote REST API and GUI documentation for e-commerce software.
Technology: Confluence, Postman, Swagger, JIRA

UBC Emerging Media Lab Vancouver, BC
Academic Assistant 11/2017 – 04/2018
Created and facilitated Virtual Reality and Brain-Computer Interface demos and workshops.
Technology: Virtual Reality (HTC Vive™), Brain-Computer Interface (Muse™ Headband)

Technical Skills

Programming Languages: MATLAB, R, Python, JavaScript, HTML/CSS, Java, C/C++
Research Software/Toolkits: PsychToolbox, EEGLAB, Python scientific computing libraries (NumPy, Pandas), E-Prime, Git, L^AT_EX, AWS (S3, EC2), Linux environments, MAXQDA
Quantitative Research Methodology: Scalp electroencephalography (EEG), event-related potential (ERP) analysis, event-related spectral perturbation (ERSP) analysis, EEG spectral parameterization, EEG transient bursts, psychophysics, eye tracking, visuomotor tracking and analysis
Qualitative Research Methodology: User interviews, usability testing, user interface wireframing and prototyping, thematic analysis

Service

CoPsy EEG Working Group, Co-Organizer 2025 – 2026
Computational Cognitive Neuroscience Journal Club, Organizer 2024
CLPS IT/Communications Committee, Twitter Manager 10/2022 – 07/2023